Introduction to Photogrammetry

Lecture 03

Process and Products

Usman Mazhar, Ph.D.

usmanmazhargeo@outlook.com

In Today's Class

1. Data Acquisition
2. Photographic Products
3. Photographic Procedures

1. Data acquisition

- Data acquisition in photogrammetry is concerned with obtaining reliable information about the properties of surfaces and objects.
- This is accomplished without physical contact with the objects which is, in essence, the most obvious difference to surveying.
- The remotely received information can be grouped into four categories

- **Geometric information** involves the spatial position and the shape of objects. It is the most important information source in photogrammetry.

- **Physical information** refers to properties of electromagnetic radiation, e.g., radiant energy, wavelength, and polarization

- **Semantic information** is related to the meaning of an image. It is usually obtained by interpreting the recorded data.

- **Temporal information** is related to the change of an object in time, usually obtained by comparing several images which were recorded at different times.

- the remotely sensed objects may range from planets to portions of the earth’s surface, to industrial parts, historical buildings or human bodies.
- The generic name for data acquisition devices is **sensor**, consisting of an optical and detector system.
- The sensor is mounted on a **platform**.
- The most typical sensors are cameras where photographic material serves as detectors.
- They are mounted on airplanes as the most common platforms.

2. Photographic Products

- Photographic products are derivatives of single photographs or composites of overlapping photographs.
- During the time of exposure, a latent image is formed which is developed to a negative. At the same time diapositives and paper prints are produced.
- A better approximation to a map are rectifications.
- A plane rectification involves just tipping and tilting the diapositive so that it will be parallel to the ground.
- Only a differentially rectified photograph, better known as orthophoto, is geometrically identical with a map.

Computational Results

- Aerial triangulation is a very successful application of photogrammetry. It delivers 3-D positions of points, measured on photographs, in a ground control coordinate system, e.g., state plane coordinate system.

Maps

- Maps are the most prominent product of photogrammetry.
- They are produced at various scales and degrees of accuracies.
- Planimetric maps contain only the horizontal position of ground features while topographic maps include elevation data.
- Thematic maps emphasize one particular feature, e.g., transportation network.

3. Photogrammetric Procedures

- Photogrammetry can be understood using a systems approach.
- The system consists of inputs, processes, and outputs.
- Typical Input: Aerial photographs.
- Typical Output: Maps, Digital Elevation Models (DEMs), or spatial data products.
- The main objective of photogrammetry is to transform photographic data into useful geographic information.

- Aerial photographs contain large amounts of raw visual data.
- Maps represent organized and interpreted spatial information.
- The photogrammetric process converts image information into cartographic data.

Key Differences Between Photographs and Maps

- Three major differences exist between photographs and maps:
  - Projection system
  - Amount of data
  - Type of information representation
- Photogrammetry addresses these differences through geometric transformations and data processing techniques.

Projection System

- Aerial photographs follow central projection (perspective projection).
- Maps use orthogonal projection (vertical projection).
- Orthogonal Projection (Map): Lines of projection are parallel and perpendicular to the map plane, meaning it eliminates distortion from perspective and terrain relief.
- Central Projection (Photograph): Lines of projection radiate from a single point (the camera lens), resulting in distortions such as perspective, tilt, and terrain relief.

Photogrammetry must convert central projection into orthogonal projection.

- Analog photogrammetry: Mechanical and optical instruments.
- Analytical photogrammetry: Computer-based mathematical calculations.

Amount of Data

- Aerial photographs contain extremely large amounts of data.
- Maps contain very small amounts of selected information.
- Example
- Photograph size: 9 inches (≈22.8 cm), Pixel size: ≈5 μm (micrometers). Total data per photograph: ≈0.5 GB
- In comparison, a map representing the same area may contain only a few kilobytes (KB) of data.

Data Reduction in Photogrammetry

- Because aerial images contain huge amounts of data:
- A key photogrammetric task is data reduction.
- Only important geographic features are extracted.
- Unnecessary visual information is discarded or simplified.
- This allows conversion of large image data into compact map data.

Information Representation

Photograph Information is implicit

- In a raw photograph, information is hidden, or implicit. You can see trees, houses, and roads, but you cannot instantly measure the exact acreage of a field, the precise height of a building, or the exact distance between two points without performing photogrammetric calculations. The spatial data (scale, orientation, coordinates) is present but hidden within the distortions of the photograph.

In maps, Information is explicit

- A map presents information explicitly. It is a processed, cartographic product where data has been corrected for distortion, scaled correctly, symbolized, and georeferenced. You can read the distance, elevation, or location directly from the map, because the data is clearly defined and interpreted.

Feature Identification and Extraction

- One major task in photogrammetry is “Feature Extraction”.
- This involves identifying objects in images, converting them into map features
- Examples: Road, networks, Rivers, Buildings, Land cover types.
- This process makes implicit image information explicit.

Thank you!